

Track 1: Agent Infra

This handbook is intended for teams that have registered or plan to register for the GOAI [Agent Infra] track. It helps participants understand the track’s focus, participation process, submission requirements for each stage, technical requirements, evaluation criteria, and support channels.

01 Track Overview

1.1 Basic Track Information

Track Name	Track 1: Agent Infra
Challenge Title	Multi-Agent Autonomous Collaboration for Complex Tasks
One-Sentence Description	Focuses on multi-agent infrastructure and collaboration systems for complex tasks, driving enterprise-level agents from demo to production.
Challenge Keywords	Multi-agent collaboration, AgentTeams, Skill, MCP, RAG, observability, zero-touch operations, Agent Infra
Challenge Co-organizers	Alibaba Cloud, DataWhale
Open-Source License Support	MCP

1.2 Track Introduction

The [Agent Infra] track focuses on multi-agent infrastructure and collaboration systems for complex enterprise-level tasks. It encourages participants to draw upon their own industry expertise, professional fields, or real-world

business scenarios to address complex, industry-wide challenges. The goal is to build reusable skills, tool integration, and operational validation capabilities, thereby advancing agents from the demo phase to production.

This track focuses not merely on demonstrating the capabilities of a single agent, but on how multiple agents can, under complex constraints, decompose tasks, pass context, invoke tools, validate results, document execution evidence, and conduct security audits—thereby forming an end-to-end task loop tailored to real-world enterprise scenarios.

1.3 Core Competition Requirements

Entries should be based on real-world enterprise scenarios and design a complete, closed-loop solution involving at least three agents with different functions, while encapsulating key capabilities as reusable skills—rather than one-time scripts or single demonstrations.

Entries should emphasize the following:

- Multi-agent collaboration to complete complex tasks;
- Skill reusability and engineering-level encapsulation;
- Stable integration with tools, systems, or cloud products;
- Context management and documentation of execution evidence;
- Security boundaries, approval workflows, rollback, and audit mechanisms;
- Plans for openness/open-source and long-term growth value of the submission.

1.4 Suitable Participating Teams

This competition is suitable for the following teams:

- Development teams with the capability to design multi-agent systems;
- Technical teams familiar with cloud products, enterprise systems, DevOps, ITSM, FinOps, or security system integration;
- Teams with an understanding of complex enterprise-level scenarios such as operations, security, cost management, and capacity management;
- Technical teams familiar with business systems, data platforms, R&D toolchains, or industry application integration;
- Teams with an understanding of challenges in real-world industry scenarios, such as complex task collaboration, process automation, knowledge retention, risk control, efficiency improvement, or user experience optimization;
- University labs, open-source teams, enterprise technical teams, or independent AI builders with capabilities in engineering implementation, toolchain integration, and demo development;
- Preference will be given to those with experience in open-source project development, technical documentation writing, or community collaboration.

02 Awards

This track offers track-specific awards and related special incentives; all prize amounts are listed before taxes.

Award	Number	Award Description
GOAI Grand Prize	1	RMB 1 million

Award	Number	Award Description
Track Champion	1	RMB 500,000
Track Runner-up	1	RMB 300,000
Third Place	1	RMB 100,000
Open Source Rising Star Award	4th–10th place, 7 teams	Gifts worth RMB 5,000 per team
Open Source Impact Award	11th–15th place, 5 teams	Gifts worth RMB 3,000 per team
Excellence Award	Teams Advancing to the Semifinals	Commemorative Gift Pack worth RMB 500 per Team
Participation Award	Teams that Submit Valid Proposals	Commemorative T-shirt per Team

03 Competition Stage Overview

Stage	Timeline	Key Milestones/Outcomes	Key Deliverables
Registration Opens	7.16	Competition Registration Begins	Registration Information
Preliminary	7.16-8.16	August 16: Deadline for preliminary submissions	Project Overview, Proposal PPT—see Preliminary Submission Guidelines for details
Preliminary Review	8.17-8.24	August 24: Announcement of Semi-Finalist Teams; Top 30 teams	No additional submissions required unless otherwise specified by the Organizing Committee

		advance to the Semi-Finals	
Semifinals	8.25-9.3	Sept. 3: Deadline for semifinal submissions	Updated project proposals, executable AgentTeams code packages, and runnable demos or demo videos—see the Semifinal Submission Guidelines for details
Semifinal Review	9.4-9.10	September 10: Announcement of the list of teams advancing to the finals; the Top 15 teams will advance to the finals and be invited to participate in the in-person final defense to compete for the final awards	No additional submissions required unless otherwise specified by the Organizing Committee

Stage	Timeline	Key Milestones/Outcomes	Key Deliverables
Finals	9.22	In-Person Final Presentation and Demonstration	Final pitch deck, live demo, final version of the code repository, or equivalent verifiable project materials. See the Final Submission Guidelines for details
GOAI DAY / Awards Ceremony	9.23	Awards Ceremony, Project Showcase, and Ecosystem Networking	Final pitch deck, live demo

*The schedule is subject to flexible adjustments based on actual progress; the Organizing Committee’s final notice shall prevail.

04 Registration Phase: Registration, Team Formation, and Topic Selection

4.1 Registration Method

Participants must register through the official website. Each team must submit:

- Registrant’s Name
- Country/Region
- Mobile Phone Number
- Primary Email Address
- Team Name
- Total Number of Team Members
- Team Members/Roles and Contact Information
- Preferred Track

4.2 Team Formation Rules

1. Participants may compete individually; teams may consist of no more than 3 members.
2. Once a team has been submitted for the preliminary, participant information may not be changed.
3. For teams advancing to the semi-finals, the Organizing Committee will verify the real-name information of all team members. If verification is not completed as required or if real-name information is found to be false, the Organizing Committee reserves the right to disqualify the relevant members or the entire team from further participation and award eligibility.

4.3 Topic Directions (Open-Ended Topics)

Participating teams may refer to the following directions when selecting their topics; however, these directions are provided solely for inspiration and do not constitute limitations on the scope of the competition. The organizers encourage participants to draw upon their own professional backgrounds, industries, research practices, corporate projects, or real-world work and study scenarios to identify complex task problems with industry-wide relevance, and to propose solutions based on the areas they are most familiar with and have the deepest insights into.

Entries are not limited to areas such as operations and maintenance, security, or cost management. As long as the topic stems from real-world needs, demonstrates the value of multi-agent systems in task decomposition, collaborative execution, tool invocation, result verification, and knowledge consolidation, and has the potential to be refined into reusable skills, tool interfaces, or open-source capabilities, it qualifies as a valid topic for this track.

The following directions serve as reference examples to help participating teams understand the track's focus and develop their proposals:

Direction 1: Zero-touch Operations

Build a multi-agent closed-loop system centered on alert aggregation, root cause identification, remediation execution, recovery verification, and incident post-mortem analysis.

Reference scenarios include:

- Multi-source alert aggregation and noise reduction;
- Automated root cause identification;
- Remediation recommendation generation and execution;
- Service recovery verification;
- Incident post-mortem and knowledge consolidation.

Direction 2: Autonomous Closed-Loop Intelligent Customer Service

Build a multi-agent closed-loop system centered on multi-channel conversation aggregation, intent recognition and prioritization, solution generation and execution, result verification and customer confirmation, as well as case review and knowledge consolidation.

Reference scenarios include:

- Multi-channel ticket/conversation aggregation and deduplication (email, online customer service, phone call transcripts, social media comments);
- Intent recognition and automatic ticket classification and prioritization;
- Resolution generation and automatic execution (refunds, exchanges, account changes, etc.);
- Verification of resolution results and confirmation of customer satisfaction;
- Review of complex cases and consolidation of customer service knowledge into a knowledge base.

Direction 3: Collaboration Throughout the Entire Software Development Lifecycle

Build a multi-agent closed-loop system centered on defect/requirement aggregation, root cause analysis of code issues, generation and execution of fixes, testing, validation, and release confirmation, as well as post-mortem analysis and knowledge consolidation.

Reference scenarios include:

- Aggregation and deduplication of defect/requirement information from multiple sources (issues, logs, user feedback);
- Automatic code defect localization and impact analysis;
- Generation of fix solutions and automated code execution;
- Test verification and canary-release result confirmation;
- Post-deployment review and documentation in the R&D knowledge base.

Direction 4: Financial Risk Control and Claims Automation

Build a multi-agent closed-loop system centered on the aggregation of multi-source risk signals, risk identification and verification, generation and execution of mitigation plans, verification of results and compliance audits, as well as post-incident reviews and the consolidation of risk control knowledge.

Reference scenarios include:

- Aggregation and noise reduction of multi-source risk signals (transaction logs, credit data, public sentiment, complaint records);
- Automated identification of root causes of risk/fraud and analysis of their scope of impact;
- Generation and automated execution of claims or credit approval recommendations;
- Response-result verification and compliance-audit confirmation;
- Risk-event review and risk-control knowledge-base capture.

05 Preliminary: Solution Design and Submission of Materials

5.1 Preliminary Objectives

The preliminary round focuses on evaluating the project's direction, technical approach, open/open-source value, and feasibility. Submitting working code is not mandatory.

5.2 Preliminary Submission Materials

Submission Materials	Required	Format Requirements	Content Description
Project Overview	Required	Written Description (500 characters or fewer)	● Project Name: Approximately 20 characters ● Problem and Scenario: What real-world problem does it solve, and for which users or scenarios is it designed? ● Core Solution: What technical approach is used for implementation, and what are the core systems, models, agents, and toolchains? ● Innovation and Differentiating Advantages: What are the key innovations or advantages compared to existing solutions? ● Open-Source / Reusability Value: Which outcomes can be made open-source, reused, or verified, and how will they be continuously iterated upon in the future? ● Current Progress: What has been completed so far? Are there any demos, prototypes, code, experimental results, or evaluation data?
Proposal PPT	Required	PPT / PDF	We recommend covering the following categories: 1. Scenario and Value: Describe the origin of the target scenario, target users, core pain points, real needs, business value, inputs and outputs, and replicability within the industry. 2. Solution Design: Present the system architecture, agent roles, task decomposition, collaboration workflows, context passing, result verification, and exception handling. 3. Skill and Tool Integration: Describe the core Skill design, reuse value, integration paths for tools, cloud products, and enterprise systems, as well as capability planning for MCP, RAG, observability, and other areas. 4. Feasibility and Implementation Plan: Outline evaluation metrics, the plan for developing a functional demo, security boundaries, risk

			control, and open/open-source plans and boundaries.
Executable AgentTeams Code Package	Optional	Repository Link / Compressed Archive	Not mandatory; if submitted, it must include an entry point for execution, dependency descriptions, configuration files, sample input/output, and proof of execution.

5.3 After Passing the Preliminary Round

Teams advancing to the semi-finals will proceed to the semi-final stage as notified by the Organizing Committee and will receive relevant materials, Q&A sessions, mentor guidance, or resource support according to the competition schedule. The specific number of qualifying teams, resource distribution methods, and optimization recommendations will be further communicated by the Organizing Committee via email, community channels, or other means.

06 Semifinal Stage: Demo Implementation and Engineering Validation

6.1 Semifinal Objectives

The semifinals will focus on evaluating project completion, demo functionality, engineering implementation, evaluation results, and open/open-source standards.

6.2 Semifinal Submission Materials

Submission Materials	Required	Format Requirements	Content Description
Updated Project Proposal	Required	PPT / PDF	Update the scenario closed-loop, architecture, Skill design, risk boundaries, and implementation plan based on feedback from the preliminary round, covering the following areas: 1. Scenario and Proposal Updates: Describe the target scenario, core pain points, business value, inputs and outputs, replicability, and major adjustments made compared to the preliminary round. 2. Complete Scenario Workflow: Demonstrate at least one complete scenario workflow that is fully operational and verifiable, including task triggering, AgentTeams orchestration, multi-agent collaboration, tool invocation, result verification, and evidence documentation. 2. Provide sample inputs and outputs, logs, traces, metrics, evaluation results, and evidence of automated verification. 3. Skills and Engineering Implementation: Describe the core Skill design, reuse methods, integration of tools / MCP / RAG / observability, interface schemas, data flows, deployment configurations, and engineering architecture. 4. Operational Validation and Risk Boundaries: Provide sample inputs and outputs, logs, traces, metrics, and evaluation results; describe failure handling, exception handling, permissions, approvals, rollback, and audit mechanisms. 5. Open / Open-Source and Implementation Plan:

			Describe the open-source license, third-party dependencies, commercial APIs, closed-source models, data licensing, compliance boundaries, and plans for ongoing maintenance and implementation evolution.
Executable AgentTeams Code Package	Required	Repository Link / Archive	Include the execution entry point, dependency descriptions, configuration files, sample inputs and outputs, and execution evidence.
Runnable Demo / Demo Video	Required	Online Experience / Local Deployment / Video Demonstration	Demonstrate at least one complete scenario workflow, verifying agent collaboration, tool invocation, output results, and exception handling. Used to showcase the complete task lifecycle, execution evidence, failure handling, and key technical highlights.

6.3 After Passing the Semifinals

Teams advancing to the finals will be invited to participate in an in-person final defense, featuring a live pitch, demo presentation, and Q&A session with experts. The Organizing Committee will announce the list of finalists, defense arrangements, on-site testing requirements, and presentation guidelines on the official website, as well as via email and community channels.

07 Final Round: On-site Pitch and Project Presentation

7.1 Final Round Objectives

The final round will focus on evaluating the effectiveness of the project presentation, on-site defense capabilities, open/open-source value, implementation potential, and long-term growth prospects.

The judges will pay particular attention to whether the project truly completes an end-to-end task cycle and whether it is operational, verifiable, auditable, and capable of sustainable evolution.

7.2 Final Round Submission Materials

Submission Materials	Required	Format Requirements	Content Description
Final Round Pitch PPT	Required	PPT / PDF	It is recommended to develop a comprehensive narrative centered on the on-site pitch, expert Q&A, and subsequent open/open-source plans, including: 1. Scenario Value: Explain the origin of the topic, target users, core pain points, real-world value, industry relevance, and potential for replication and scaling. 2. Solution and Technical Highlights: Showcase the system architecture, AgentTeams orchestration, multi-Agent task division and collaboration, core Skills, tool integration, data flows, and key innovations. 3. Operational Results and Validation Evidence: Demonstrate demo running results, end-to-end task closed-loop processes, sample inputs and outputs, logs/traces/metrics, evaluation results, and exception handling effectiveness. 4. Engineering Maturity and Security Boundaries:

			Describe deployment methods, dependency disclosure, access controls, approval, rollback, auditing, degradation and reproducibility. 5. Supporting Materials for the Defense: Provide supplementary details on architectural specifics, technical trade-offs, resource costs, evaluation methods, failure cases, and risk mitigation strategies that judges may inquire about. 6. Open / Open-Source and Long-Term Plans: Describe the scope of open-source code, license selection, Skill distribution, version iterations, community operations, industry/research collaboration, and long-term maintenance plans.
On-site Demo	Required	On-site Demo / Online Experience / Local Deployment	Demonstrate at least one complete scenario workflow that verifies agent collaboration, tool invocation, output results, and exception handling. Used to showcase the complete task lifecycle, execution evidence, failure handling, and key technical highlights. ● You may use the Element chat room built into AgentTeams. ● You may also use a custom-developed WebUI or other visualization solutions.

Submission Materials	Required	Format Requirements	Content Description
Final version of the code repository or equivalent verifiable project materials	Required	Repository Link / Review-Ready Delivery Package	Provide the final, review-ready version, including a README, deployment instructions, open-source license, sample configurations, and testing methods.
Industry / Research Collaboration Needs	If applicable	Form	If a contestant has commercialization needs, please specify the industry scenarios, corporate partners, research institutions, investors, or open-source communities they wish to collaborate with; the Organizing Committee will provide links to venture capital resources. If none, leave blank

08 Technical Requirements for Entries

This chapter outlines general technical requirements applicable across all stages. During the preliminary round, participants may focus on explaining their design concepts; in the semi-finals and finals, they must progressively develop executable and verifiable engineering deliverables.

8.1 Multi-Agent Collaboration Requirements

- The submitted proposal must design at least 3 Agents with distinct functions. Each Agent must have a clearly defined identity and collaborate to complete an end-to-end closed-loop task.
- The multi-Agent design must be based on the AgentTeams framework as the foundation for collaborative design, explaining how role orchestration, task decomposition, context passing, collaborative execution, and state tracking map to the capabilities of this framework.

8.2 Agent Identity List

Participating teams must submit an Agent Identity List to describe the identity attributes, capability boundaries, and collaborative relationships of each Agent in the multi-Agent system; see Appendix A for details.

8.3 Multi-Agent Closed-Loop Description

The competition proposal must describe how the multi-Agent system completes the following closed loops:

1. **Task Input:** The system receives alerts, tickets, logs, invoices, security incidents, or other enterprise task inputs.

2. **Task Decomposition:** The master Agent or collaboration framework decomposes the task and assigns it to Agents with different functionalities.
3. **Context Passing:** Task context, historical information, tool execution results, and interim conclusions are passed between different Agents.
4. **Tool Invocation:** Agents invoke Skills, MCP tools, cloud products, enterprise systems, knowledge bases, or external APIs to complete tasks.
5. **Result Verification:** The system verifies results such as remediation, security responses, and cost optimization.
6. **Execution Evidence Documentation:** The system preserves logs, traces, metrics, reports, screenshots, or other evidence of execution.
7. **Approval and Rollback:** High-risk actions must include mechanisms for manual confirmation, approval, rollback, and auditing.
8. **Knowledge Consolidation:** Execution results, post-incident reviews, or experience-based rules are consolidated into reusable capabilities.

09 Skills, MCP, RAG, and Observability Requirements

9.1 Skill Requirements

Skills are a mandatory component of this competition.

Entries may use official Alibaba Cloud Skills or encapsulate key capabilities as reusable Skills. Skills should serve as an abstraction layer for task capabilities, rather than merely describing one-time agent behaviors.

Each proposal must provide at least a list of core Skills, specifying:

- Skill name;
- Skill purpose;
- Inputs and outputs;
- Invocation conditions;
- Dependent tools;
- Failure handling mechanism;
- Security boundaries;
- Reusability;
- Relationship with multi-agent collaboration workflows.

9.2 MCP and Tool Integration Requirements

MCP is a recommended **optional component for this competition**.

MCP is the recommended protocol for integrating external tools and systems, used to connect to external tools such as cloud products, enterprise systems, databases, knowledge bases, ticket systems, CI/CD, and monitoring systems.

Skill takes the task capability abstraction layer and MCP takes the tool connection layer.

If MCP is not used in the scheme, an equivalent external tool integration contract shall be given, including:

- Agreement;
- Authentication mode;
- Input and output Schema; ;
- Error handling;
- Audit records;
- Cost of subsequent migration to MCP.

The "equivalent integration contract" here means that even if MCP is not adopted for the time being, the external tools need to be abstracted into tool capabilities that can be stably called by Agent/Skill, and the tool name, call

entry, parameter Schema, return structure, permission range, failure retry, idempotent control, audit log and degradation mode need to be explained.

During the review, it is not required that the alternative has implemented MCP Server, but it needs to be able to judge whether it only needs protocol adaptation when it is migrated to MCP, rather than redesigning the tool call chain.

9.3 Observability Requirements

Observability is the recommended **option in this competition**.

Observability is the recommended but not mandatory key technology of multi-agent system from Demo to Production. It is suggested to cover the whole link reasoning trajectory such as Skill call, MCP tool, RAG retrieval, LLM reasoning, and support scene practice such as online monitoring and alarm, evaluation and optimization.

If the contestant uses the observability, it should explain:

- Observation data collection method;
- Semantic specifications, such as whether to follow the OpenTelemetry GenAI standard;

- Data type;
- Back-end storage and retrieval scheme;
- Application scenarios and effects.

Observable system is recommended to cover at least the following 1-2 types of data:

1. Trace

Fully record Span of Agent/Skill/MCP/RAG/LLM, and try to follow the open-source semantic standards.

2. Log

Structurally record the decision-making basis, failure reasons, authority check and approval events, and suggest to associate with Trace through Traceld.

3. Metrics

Cover core indicators such as traffic, performance, cost and reliability, such as the number of conversations/sessions, end-to-end delay/TTFT, Token consumption and Tool success rate.

It is recommended to support real-time or offline evaluation built on observational data to scientifically quantify the performance and efficiency of agent reasoning. Evaluation methods can include rule evaluation, LLM-as-Judge, Agent-as-Judge or manual marking. Evaluation results may be fed back into datasets, or leveraged by specialists to tune agent Prompts, Skills, RAG modules, MCP and large language models (LLMs).

9.4 RAG and Context Enhancement Requirements

RAG is a recommended **optional component for this competition**.

RAG is a recommended but not mandatory context enhancement technology, which is suitable for schemes that need to retrieve knowledge base, history, standards and specifications, business data, Runbook, review evidence or tool execution results.

RAG should be used as a contextual ability in the call chain of Agent, Skill and MCP; MCP accesses the data source, Skill encapsulates retrieval, evidence alignment and result writing, and Agent judges whether the retrieval result is enough to support decision-making.

According to the actual situation, the entry scheme should achieve at least two of the following four abilities:

1. Agent memory storage

Store Agent session history and decision context, and support time window query and vector semantic retrieval.

2. Knowledge base RAG

For static external knowledge, such as specification, Runbook, business data, etc., document vectorization and similarity retrieval are realized based on various vector databases.

3. Shared state management

Manage the cooperation state of multi-agent to ensure the consistency under concurrency.

4. Trace Observability

Persist the execution trace and evidence chain into various storage engines to support full link playback and audit.

If a proposal explicitly does not use RAG, it must implement at least two of the remaining three items (excluding “Knowledge Base RAG”) and demonstrate the effectiveness of its context mechanism.

10. Recommended Toolchains and Resource Usage Guidelines

Entries must specify the open-source projects or cloud products used, their versions or compatibility scope, methods of invocation, and their relationship with the Agent, Skill, MCP, and RAG.

The preliminary round focuses primarily on proposal design; submission of executable code is not required. However, designs involving multi-Agent collaboration must be based on AgentTeams (formerly known as Hiclaw), and must explain how role orchestration, task decomposition, context passing, collaborative execution, and state tracking map to the capabilities of this framework.

Technical Focus	Open-Source Projects/Tools	Requirements	Official Website	Configuration
Multi-Agent Collaboration Framework	AgentTeams (formerly Hiclaw)	Required	hiclaw.io	As the design foundation for multi-agent collaboration solutions, this section explains how role orchestration, task decomposition, context passing, collaborative execution, and status tracking map to the framework's capabilities.
Cloud Skills	Cloud Skills Portal	Recommended	skills.aliyun.com	As the fundamental means for agents to interact with the cloud, this section focuses on handling and addressing authentication, orchestration, and end-to-end experience during the use of Skills. This helps contestants better focus on building Skills specific to their project, eliminating the need to build additional Skills for foundational capabilities.

AI Management Center	Nacos	Recommended	nacos.io	Provide management capabilities for AI applications, Agents, Skills, prompts, configurations, service discovery, and runtime governance, with a focus on governance design and interface relationships.
AI Gateway	Higress	Recommended	higress.io	Provide a unified entry point, authentication, routing, rate limiting, and observability for model services, Agent services, or external tool calls, with a focus on gateway governance policies.
Agent Data Layer	PolarDB for PostgreSQL	Recommended	PolarDB-for-PostgreSQL	Serve as an optional implementation for storage systems such as vector storage, long-term memory, RAG, and audit logs, with a focus on data models, index design, permission boundaries, and replaceability.
	UnifiedModel	Recommended	alibaba.github.io/UnifiedModel/	Enable consistent modeling, association, and querying of data across different systems by providing a unified description of entities, data, relationships, and storage.

Technical Focus	Open-Source Projects/Tools	Requirements	Official Website	Configuration
Message Queues	RocketMQ	Recommended	rocketmq	It can be used for event-driven, asynchronous task scheduling, message collaboration among Agents, execution state flow and reliable notification, with a focus on reliability and event model.
Observability	Open source LoongSuite+AgentScope Studio or Alibaba Cloud AgentLoop	Recommended	oongsuite agentscope-studioA libaba Cloud AgentLoop	It can be used to record the reasoning trajectory of Agent as data support to improve the multi-dimensional indicators such as agent quality, performance, cost and reliability.

Tool chain description

- Required: AgentTeams.
- Recommended: Alibaba Cloud official cloud Skills, Nacos, Higress, PolarDB for PostgreSQL, RocketMQ, LoongSuite, AgentScope Studio and AgentLoop.
- Alternatives can be used, but the interface compatibility, replacement reason and migration cost should be explained.
- Recommended projects and cloud products are not scored according to the number of uses. The focus of the review is not on "stacking tools", but on whether the design concept, interface contract, necessity, substitutability, authority boundary, end-to-end closed-loop evidence and subsequent migration costs are clearly stated in the entry scheme.

11 Open Source and Compliance Requirements

GOAI encourages the participating teams to participate in the competition under the principles of openness, sharing and co-creation, and the participating works to form reusable results in terms of codes, documents, tool components, Skill, interface contracts, sample schemes and operation reports.

This track does not require all works to be completely open source in the semi-final or final rounds, but the participating teams need to clearly explain:

- Project open source or open scope;
- Open source agreement or authorization method;
- Third-party dependence;
- Call of commercial API;
- Usage of closed-source model;

- Data source and authorization boundary;
- Reproducible mode;
- Deployment dependency;
- Follow-up maintenance plan.

11.1 Specific requirements

Project	Requirements statement
Is open source code required?	It is not mandatory to submit the code in the preliminary round, but it is necessary to explain the open source plan and boundary; After entering the semi-finals/finals, the runnable AgentTeams code or equivalent verifiable engineering materials shall be submitted, and the repository or delivery package accessible for review shall be provided.
Are deployment instructions required?	Yes. Starting from the semi-final, it is necessary to provide the operation entrance, dependent installation, configuration method, environmental requirements and description of repeated experiments.
Are commercial APIs allowed?	Yes, provided that you fully disclose the call link, cost assumption, scope of authority, substitutability and potential lock-in risk.
Are closed source models allowed?	Yes, but the scope of use, reasons, alternatives, migration costs and the impact on reproducibility must be stated.
Is it allowed to use existing projects to continue development?	Yes, but the original project source, team contribution scope, new innovations and compatibility with the original agreement are required.
Data sources and authorization requirements	When using enterprise data, logs, tickets, knowledge bases or business data, the source, authorization status, desensitization method and compliance boundary must be stated.
Disclosure Requirements for Third-Party Dependencies	Open source dependencies, commercial services, MCP/tool interfaces, external system permissions, licenses and key version information need to be fully disclosed.

Other restrictions	The scheme contains at least three Agents with different functions; Skill is a required option; High-risk actions must keep the boundaries of approval, rollback and audit; When MCP is not used, the equivalent integration contract should be stated.
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Entries must clearly disclose open source protocols, third-party dependencies, commercial APIs, closed source models, data sources and authorization boundaries. High-risk execution actions must retain the ability of approval, rollback and audit; If MCP or RAG is not used, the equivalent contract or alternative and the migration cost shall be stated.

12 Evaluation criteria

12.1 General review dimensions

Review dimension	Weight	Description
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Review dimension	Weight	Description
Scenario Value and Industry Reproducibility	25%	Whether to target real, clear and representative scenes; Whether it can clearly explain the target users, core pain points, actual needs and value benefits; Whether the scheme has perceptible value in terms of efficiency improvement, cost reduction, quality improvement, experience optimization, risk control, knowledge accumulation or management innovation; Whether the related problems, scenarios or solutions are universal, and whether they can be replicated, migrated or popularized in similar industries, organizations, business processes or user groups.
Multi-Agent collaboration and autonomous closed-loop ability	25%	Whether the Agent's responsibilities are clear; Whether the task disassembly is reasonable; Whether the context transfer, cooperation protocol and state flow between Agents are structured; Whether it can handle exceptions, conflicts and multi-scheme selection; Whether there are boundaries of manual confirmation, approval, rollback and audit for high-risk actions.
Skill engineering system and ecological reuse	25%	Whether Skill covers critical mission capabilities; Whether the input/output, call conditions, dependent tools and failure handling are clear; Whether it can be reused by multiple Agents or multiple scenarios; Whether to consider version, release, rollback and quality evaluation; Whether to use AgentTeams as the collaborative base point and reasonably use the official cloud Skills in Alibaba Cloud.
Project Landing, Operation Verification and Safety Auditability	20%	Whether Demo or engineering materials can be run; Whether the deployment, configuration, model and tool dependence are clear; Whether to provide evidence such as logs, Trace, Metrics and operation reports; Whether MCP/RAG/observable link has verifiable design; Whether the data, authority, key, approval, rollback, downgrade and audit mechanism are complete.
Open/open-source contribution	5%	Whether reusable results, interface contracts, documents and examples are formed; Are open-source agreements and third-party dependencies clear?

12.2 Supplementary Customized Evaluation Criteria for This Track

1. AgentTeams will not be evaluated solely based on a simple mention of the tool name.

AgentTeams is the basic point of multi-agent collaborative design, and it should focus on verifying how role orchestration, task decomposition, context passing, collaborative execution, and status tracking map to the framework's capabilities.

2. Skill is a required item in this competition.

The review will focus on verifying Skill's input and output, call conditions, dependent tools, failure handling, verification methods, reuse value, version evolution and open-source distribution design.

3. Recommended projects and cloud products are not scored according to the number of uses.

The review focuses on the necessity, interface contract, substitutability, authority boundary, end-to-end closed-loop evidence and subsequent migration costs.

13 Red Line and Serious Deduction Items

1. The Organizing Committee has the right to cancel the qualification and results of any player who cheats during the competition once it is verified. All employees of Alibaba Cloud — the host of this competition, as well as any individuals who may have access to the background operations, products or data related to the competition challenges, are automatically disqualified from this competition and shall forfeit their eligibility to participate.

2. It is not allowed to share code with each other privately. If sharing is really necessary, the teams should be merged;

If the codes are found to be the same, the organizing Committee has the right to cancel the relevant team results.

3. Teams that only submit concept notes, PPT or marketing materials, without providing PoC, experiments, simulations, logs, videos, platform achievements or equivalent verifiable materials, shall in principle face disqualification or heavy point deductions.

4. After the final edition is sealed, if the acceptance personnel can't reproduce the core Demo, experiment or field task according to the requirements of README, deployment instructions or track platform, the team's eligibility for awards may be revoked.

5. If the final presentation significantly deviates from the commitments made in the semi-final round, consists of material hastily assembled by others rather than the team's own work, or fails to disclose the foundation of an existing project, the team's results may be disqualified.

6. Acts such as plagiarism, using pre-built projects, fabricating experimental results, manipulating community data, or purchasing fake Stars, Forks, Issues, or Pull Requests will result in the cancellation of grades once confirmed.

7. Misrepresentation regarding data, licenses, commercial APIs, closed-source models, personal information, corporate data, hardware security, or on-site control boundaries will result in point deductions, disqualification, or liability, depending on the severity of the violation.

14 FAQ

Q1: What types of teams are eligible to participate in this competition?

This competition is open to developers, university labs, open-source teams, corporate technical teams, or independent AI builders with expertise in multi-agent system design, engineering implementation, toolchain integration, or an understanding of real-world industry scenarios. Participating teams may draw upon their familiarity with specific industries, professional fields, research practices, or business scenarios to design solutions for complex, universal tasks. Solutions are not limited to the example areas of operations, security, or cost management.

Q2: Is open-sourcing the code mandatory?

The competition encourages participating teams to open-source their code, documentation, tooling components, Skills, interface contracts, sample solutions, and other deliverables; however, it does not require that all entries be fully open-sourced. Submitting code is not mandatory for the preliminary round, but teams must outline their open-source plan and its scope. Upon advancing to the semifinals and finals, teams must submit executable code or equivalent, verifiable project materials, and provide the judges with access to the repository or delivery package.

Q3: Can commercial APIs be used?

Yes, but you must disclose the call details, cost assumptions, scope of permissions, alternative solutions, and the impact on the reproducibility of the solution.

Q4: Can a closed source model be used?

Yes, but it must explain the scope of use, reasons, alternatives, migration costs and the impact on reproducibility.

Q5: Is it possible to continue development based on an existing project?

Yes, but you must specify the source of the original project, the scope of the team's contributions, the new innovations, and compatibility with the original license.

Q6: Is it mandatory to use a specific platform or toolchain?

This track requires AgentTeams as the basic point of multi-agent design, and Alibaba Cloud official cloud Skills is used. MCP, RAG, observability, etc. are recommended capabilities, and equivalent schemes can be used, but compatibility and migration costs should be stated.

Q7: What level of detail is required for the demo?

The preliminary round focuses on verifying the completeness of the proposal; feasibility may be demonstrated using prototypes, screenshots, logs, traces, or videos. For the semi-finals and finals, participants must provide a working demo or equivalent verifiable engineering materials, evidence of execution, and descriptions of failure handling and security boundaries.

Q8: How can participants from overseas receive competition notifications and answers to questions about the competition problems?

Contestants from mainland China may join the DingTalk Q&A group for the competition. Overseas contestants will receive answers to questions about the competition, access to downloadable materials, and competition announcements via the official email or international communication channels to be announced at a later date.

15 FAQs and Contact Information

15.1 Track Q&A Groups

Track 1: Agent Infra Contestant Communication DingTalk Group.

DingTalk Group ID: 186080014742



The group will provide updates on problem solutions, mentor Q&A sessions, contest rewards, resource requests, submission requirements, and other relevant information. Contestants are welcome to join the group to interact with mentors and other participants.

15.2 Communication Channels for International Participants

International participants who are unable to use DingTalk may access event notifications, download materials, and receive answers to questions about the competition problems via the official email address or other international communication channels to be announced at a later date.

15.3 Supporting Learning and Community Activities

During the competition, the Organizing Committee will organize supporting activities—such as problem briefings, online Q&A sessions, AI summer camps, and technical sharing sessions—in accordance with the competition schedule to help participating teams understand the problem requirements, how to use the toolchain, and the submission guidelines.

Specific activities, participation methods, study materials, certificates or souvenirs will be notified via official website, official communities and mail.

Note: Participation in related activities is not a prerequisite for the evaluation of contest entries; the final evaluation will still be based on the quality of the work, the submitted materials, and the judging criteria.

16 Appendix: Submission Templates

Appendix A: Agent Identity Checklist Template

Field	Filling Instructions
Name	The Agent's unique identifier.
Role	The role of an Agent.
Capabilities	What the Agent can and cannot do.
Inputs	Input information required.
Outputs	Output Format and Quality Requirements.
Dependencies	Other agents, skills, or tools on which it depends.
Decision Boundary	The boundary between autonomous decisions and those requiring human confirmation.
Trace	How the execution process is recorded, replayed, and audited.

Appendix B: Skill Checklist Template

Field	Filling Instructions
Skill Name	The name or code of the Skill.
Skill Type	Official Cloud Skill / Custom Skill / External Tool Wrapper / Enterprise System Integration Capability, etc.
Use Case	Describes the types of tasks or processes for which this Skill is suitable.
Input Parameters	Describes the inputs required to invoke this Skill.
Output Results	Describes the structured or unstructured results output by this Skill.
Invocation Conditions	Describes the conditions under which the Skill is triggered.
Dependent Tools / Systems	Describes the Cloud products, enterprise systems, MCP tools, databases, or external APIs on which this Skill depends.

Field	Filling Instructions
Failure Handling	Describes how to handle call failures, exceptional results, or timeouts.
Permissions and Security	Describes permission controls, key management, and risk boundaries during the call process.
Reusability	Describes whether this Skill can be reused in other Agents or scenarios.

Appendix C: Suggested Structure for the One-Page Project Proposal

The one-page project proposal should include:

- Project Name;
- Team Name and Member Information;
- Target Scenario and Core Problem;
- Solution Overview;
- Multi-Agent Architecture;

- Key Skills and Toolchain;
- Demo Results;
- Open Access / Open Source Contributions;
- Business Value and Scalability;
- Future Plans and Integration Requirements.

To help participants better understand the competition requirements and submit their entries effectively, the following is a sample demo for this track for reference.

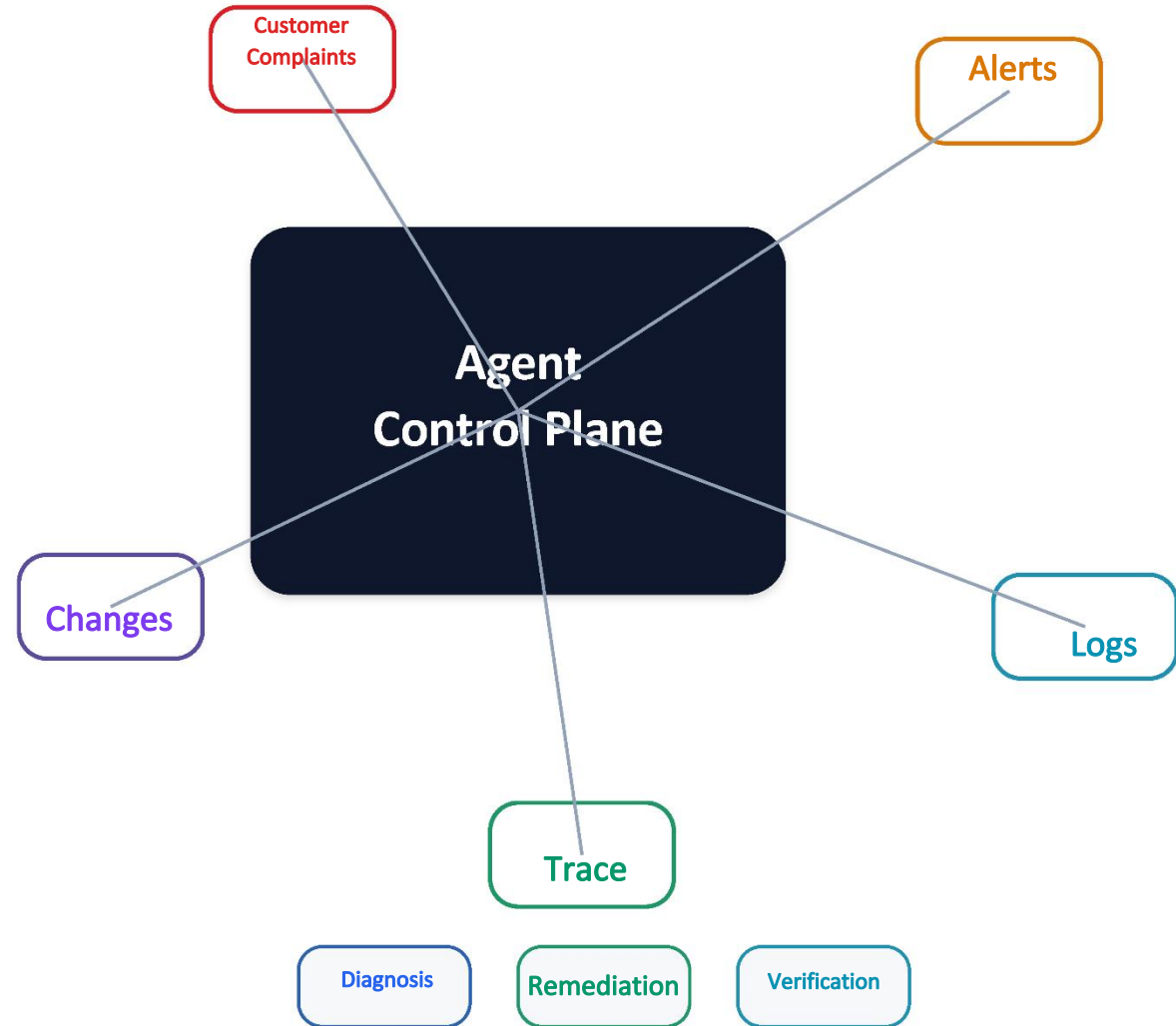
Demo Name:

OpsPilot Zero: A zero-touch multi-agent system for incident diagnosis and automated remediation across cloud and on-premises IDC environments

OpsPilot Zero

A zero-touch multi-agent system for incident diagnosis and automated remediation across cloud and on-premises IDC environments

Incident Remediation Team · Zhang San/Li Si/Wang Wu · Preliminary Round V0.2



A 24/7 Troubleshooting Agent for Every Business Team

This page brings together the scenario, key challenges, and value proposition for the preliminary round: why the project matters and how it will be implemented as a working demo.

Challenge Focus

- Multi-agent collaboration
- Skill Engineering
- Tool integration and auditing
- Reusable open-source deliverables

Aligned with the Challenge Track

Real-World Challenges

- Vague customer complaint descriptions
- Fragmented or missing data
- Significant differences across environments
- Expert knowledge that is difficult to scale

Addressing Real-World Challenges

Project Value

- Initial diagnosis within 5-10 minutes
- Controlled automated remediation for low-risk actions
- Identification of evidence gaps
- Accumulation of runbooks/Skills

Delivering Measurable Value

30-60min

Typical time required for initial manual troubleshooting

5-10min

Target time to initial diagnosis in the demo

0

Human intervention required for low-risk actions

100%

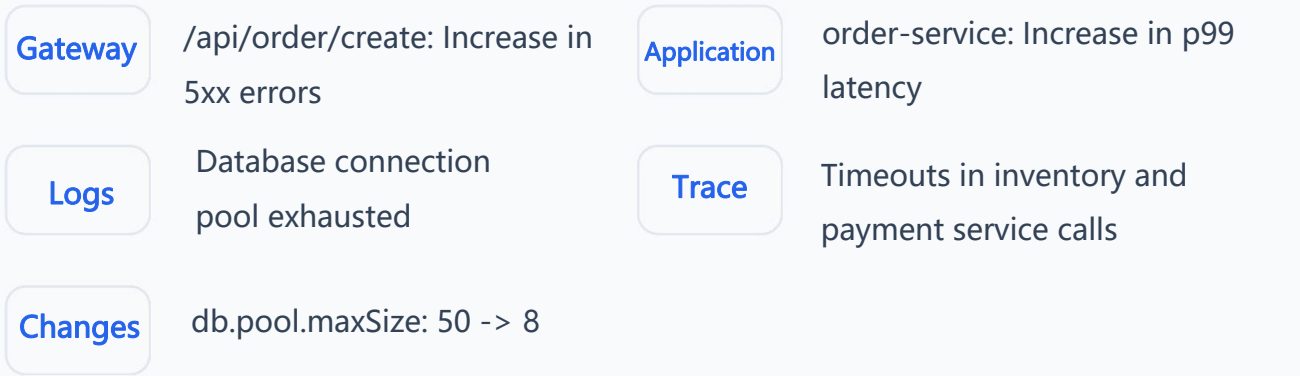
Audit logging for all critical operations

Demo Scenario: Diagnosing and Remediating an Order Service Incident

The demo simulates a customer complaint from an e-commerce company in Hangzhou and shows how a vague problem description is converted into a closed-loop automated diagnosis and remediation workflow.

Customer Complaint

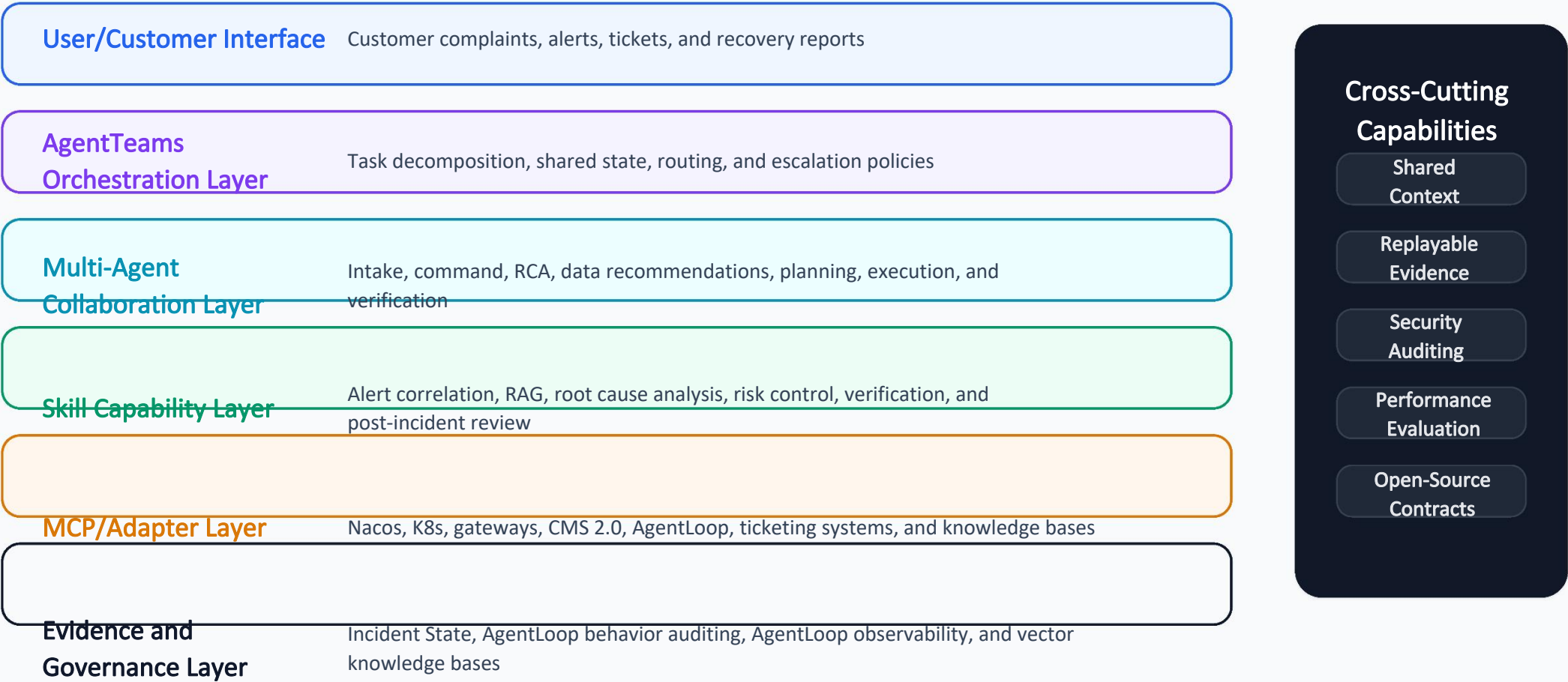
“Since 10:15, the order placement failure rate has increased, and the payment page keeps loading without completing. Please determine whether the issue is related to the application, configuration, database, gateway, or infrastructure.”



Kubernetes/ACK/on-premises clusters + Nacos +
Higress/Ingress + MySQL/RDS

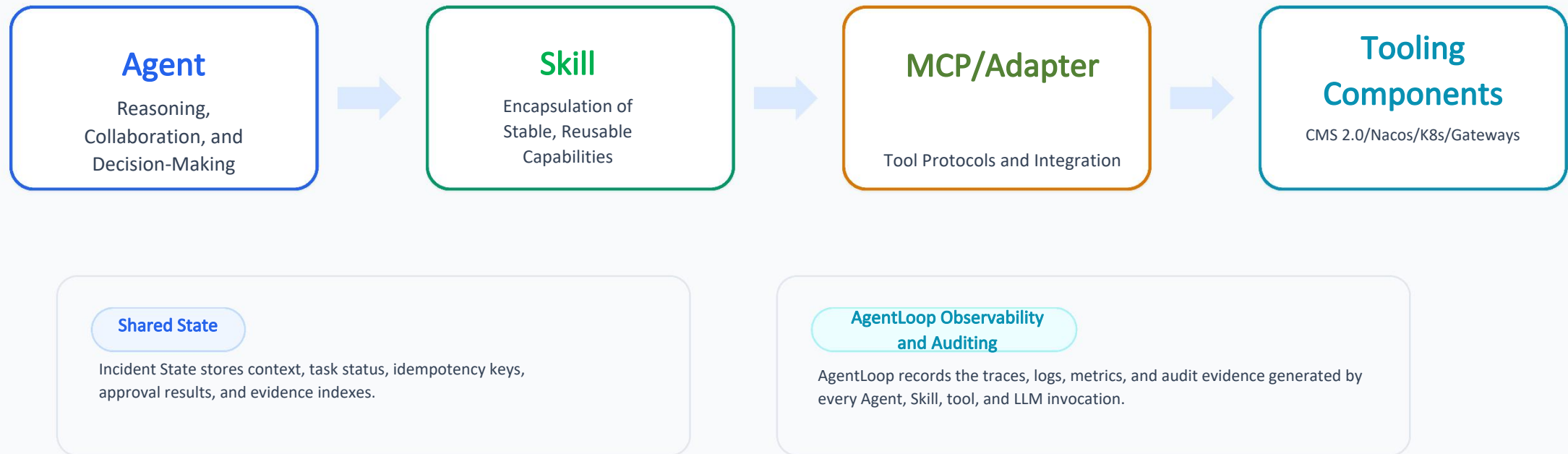
Overall Technical Architecture: AgentTeams + Skills + MCP/Adapters

Decision-making and collaboration, reusable capabilities, external tools, and evidence governance are separated into clearly defined layers.



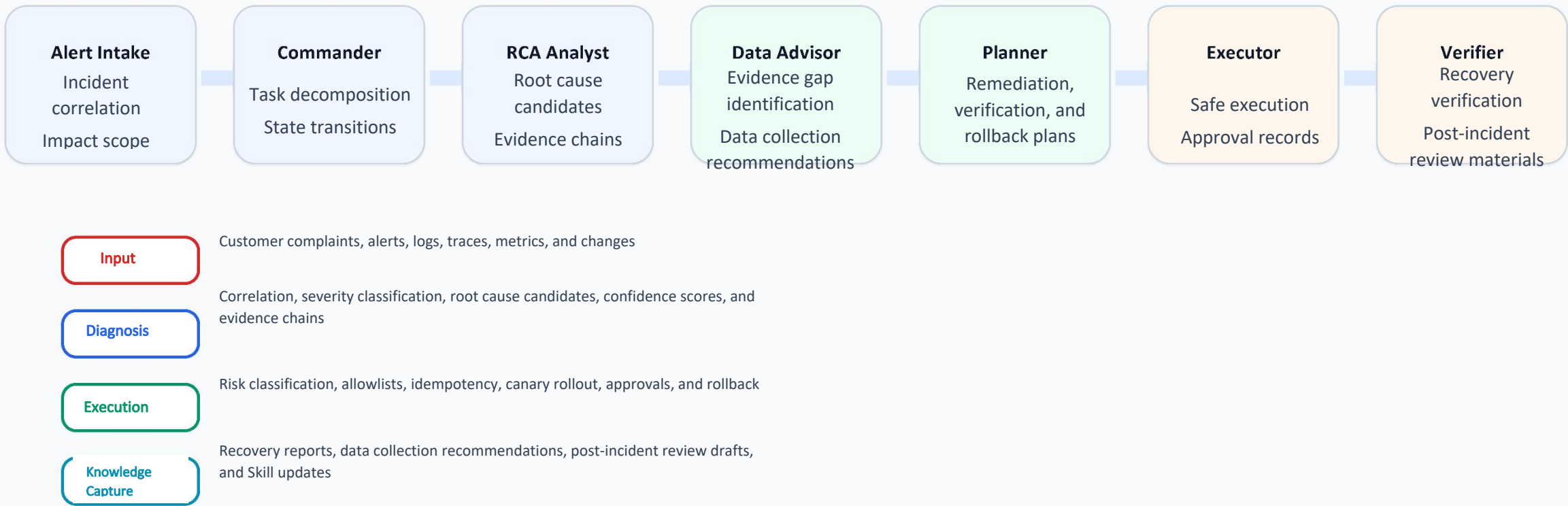
Interactions Between Agents and Tooling Components

Agents do not invoke tools directly. Instead, Skills and MCP/adapters create a governed tool invocation chain.



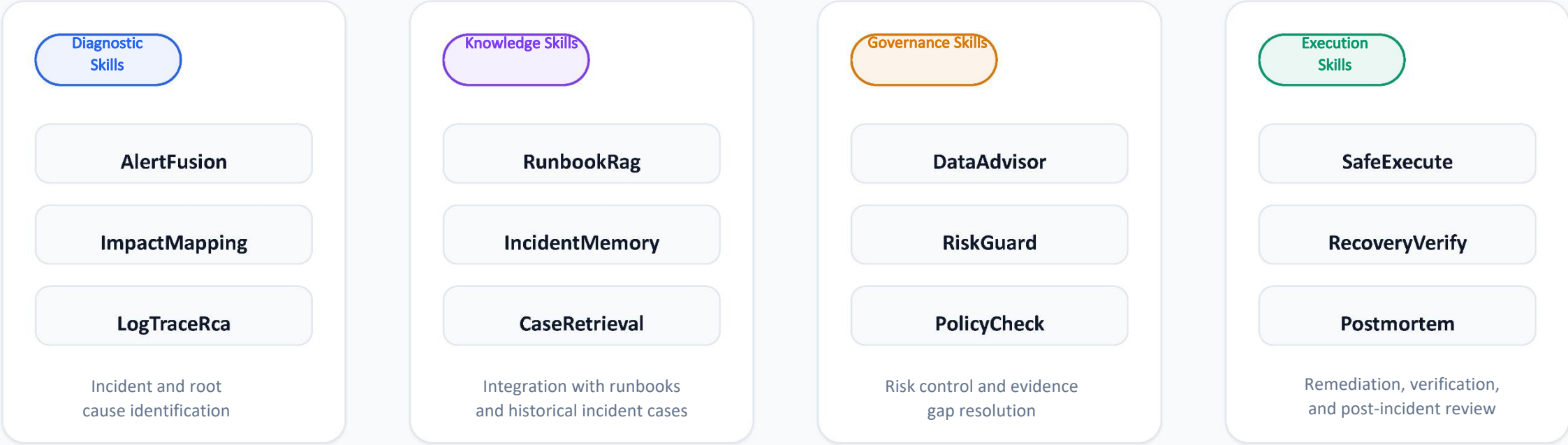
Seven Functional Agents Cover Diagnosis, Execution, Verification, and Knowledge Capture

The manual troubleshooting process is decomposed into collaborative units that can operate in parallel and remain traceable and replayable.



Core Skill System: Turning Expert Knowledge into Reusable Capabilities

Skills are the project’s core units of engineering reuse, rather than one-off prompts.



Standardized Skill Inputs and Outputs: Schemas, evidence references, failure handling, safety boundaries, evaluation examples, and open-source documentation

Tool and Infrastructure Integration: Abstracting Environmental Differences Through Standard Contracts

The preliminary round provides MCP/adaptor interfaces and mock data. Real systems will be integrated in the final round.

AgentTeams

Task decomposition, state transitions, and collaboration orchestration

Nacos

Configuration retrieval, change detection, and allowlisted rollback

Higress/Ingress

5xx errors, routing, rate limiting, and traffic management

K8s/ACK/IDC

Instance management, scaling, restarts, and canary rollout recommendations

AgentLoop

Traces, logs, metrics, alerts, and business-level evaluation

pgvector + Knowledge Base

Runbooks, historical incidents, and Agent memory

Design Principle: Skills abstract reusable capabilities, while MCP/adapters connect those capabilities to real tools. Mock and production integrations use the same schemas.

Nacos as the Governance Control Plane for Agent AI Resources

Nacos centrally manages prompts, Skills, AgentSpecs, and AgentTeams definitions, supporting review, canary release, rollback, and auditing.

AgentTeams Spec

Defines multi-agent roles, topology, collaboration events, and input/output contracts

AgentSpec Registry

Defines the Prompt, Skills, MCP integrations, Memory, and Policy for each Agent

Prompt Registry

Manages system prompts, scenario templates, variable schemas, and tags

Skill Registry

Manages Skill catalogs, version lifecycles, security reviews, and rollbacks

Release and Access Governance

RBAC, namespaces, visibility scopes, and tag-based canary releases

Runtime Discovery and Auditing

Loads resources by tag and reports invoked versions and supporting evidence

**Governance Workflow: Define Spec -> Security Review -> Tag-Based Release -> Runtime Loading
-> Invocation Audit -> Rapid Rollback**

RAG, Context, and Evidence Management

When data is incomplete, the system does not fabricate conclusions. Instead, it reports evidence strength, evidence gaps, and recommended data collection actions.



Strong Evidence	The configuration change coincides with the incident window, and the logs show that the connection pool was exhausted.
Weak Evidence	Some traces show timeouts, but the sampling coverage is insufficient to represent the full call chain.
Missing Evidence	Database slow-query logs, connection pool utilization, and instance-level JVM metrics are unavailable.

AgentLoop Observability, Evaluation, and Safety Boundaries

The key to zero-touch operations is not unlimited automation, but verifiable, auditable, and reversible automation.

AgentLoop Trajectory Tracking

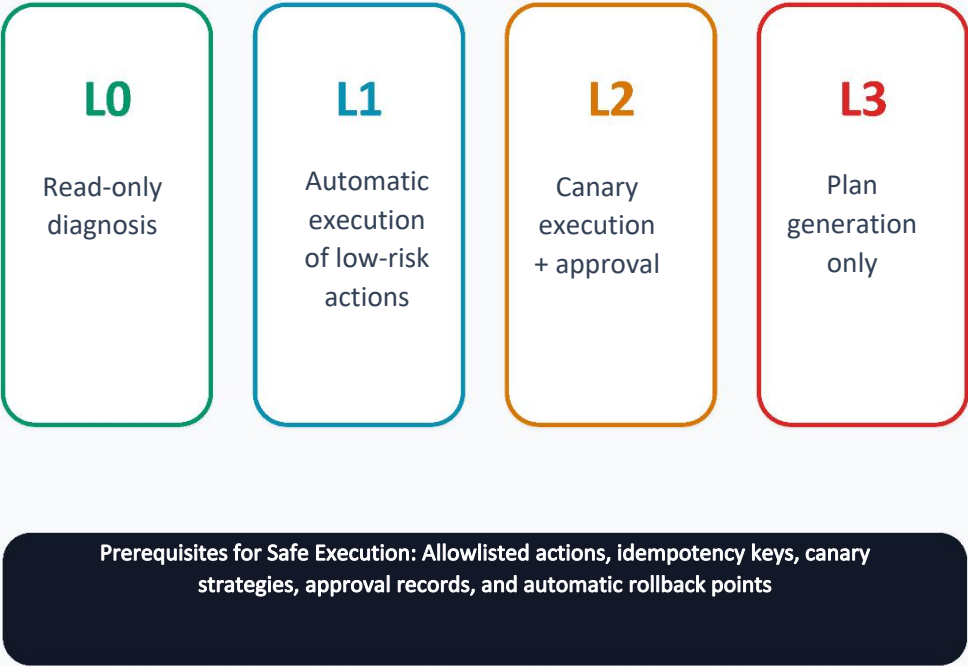
Every task decomposition step, Skill invocation, and tool invocation can be replayed.

Performance Metrics

Semantic evaluation of task completion, tool selection quality, RAG retrieval relevance, and other effectiveness indicators

Experimental Evaluation

Incident cases are replayed using Golden/Bad Case datasets to compare the performance of different Agent versions.



Demo Scope and Final-Round Roadmap

The preliminary round will deliver a credible demo. The final round will focus on production integration, evaluation datasets, and open-source engineering.

Preliminary Round V0.2

- Mock incident data and tool interfaces
- A collaborative workflow involving seven Agents
- Core Skill schemas
- Root cause analysis, remediation plans, and verification reports

Demo-Ready

Final Round V0.5

- Integration with production Nacos, K8s, and CMS 2.0 systems
- Introduction of evaluation datasets
- Allowlisted safe execution
- AgentLoop observability trace dashboard

Verifiable

Open-Source Plan

- Agent and Skill templates
- MCP adapter examples
- Incident replay datasets
- Deployment and demo scripts

Reusable

OpsPilot Zero

Transforming complex incident troubleshooting from a process dependent on individual expert experience into collaborative, governable, and reusable Agent Infra.

01

Designed for Industries of Every Kind

Supports diverse troubleshooting scenarios across public cloud, private cloud, hybrid cloud, and on-premises IDC environments.

02

Closed-Loop Multi-Agent Operations

Covers the full workflow from customer complaint correlation and root cause identification to remediation planning, recovery verification, and post-incident knowledge capture.

03

Skill Engineering

Encapsulates expert knowledge into stable capabilities that can be evaluated, reused, and open-sourced.

04

Secure and Auditable

Low-risk actions are completed automatically in a closed loop, while high-risk actions generate replayable plans that are subject to approval.

Next Steps: Submit the preliminary-round demo, complete the tool adapters, and build an incident replay evaluation dataset.

